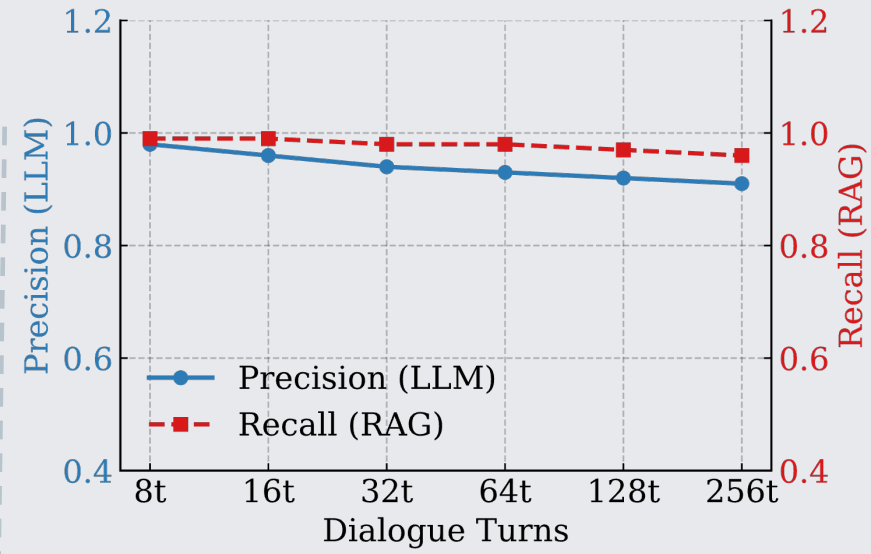
1.Object Memory

Definition: Storing and recalling explicit facts about objects.

Case:
Cue: I put my bike key in the drawer.
Query: Where is my bike key?



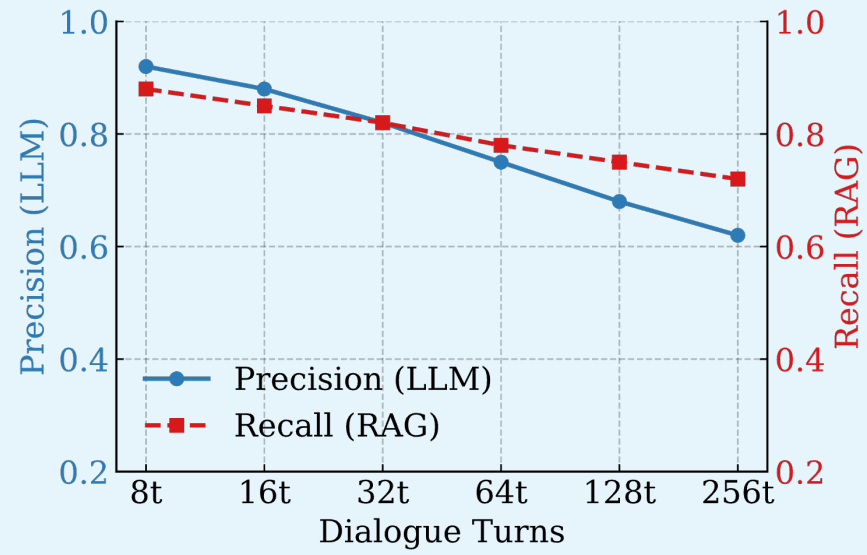
PDR = 92.85% **RTR = 96%**

Performance: **Robust & Stable**; sustains high precision across all turns.

2.Episodic Memory

Definition: Storing and recalling information about past events.

Case:
Cue: Celebrating my son's birthday.
Query: Whose birthday was it?



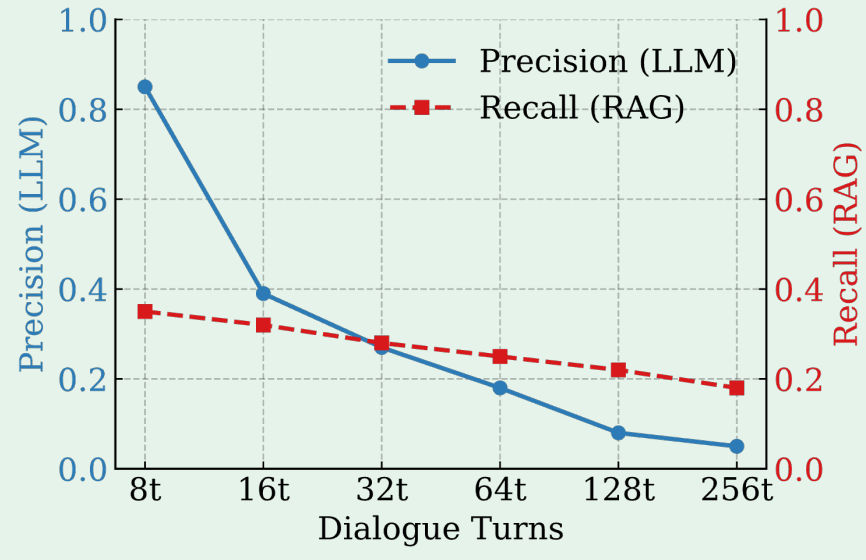
PDR = 67.39% **RTR = 72%**

Performance: **Linear Decay**; information density thins out over time.

3.Cognitive Memory

Definition: Implicit information that emerges through interaction and characterizes the user's situation.

Case:
Cue: I'm on a diet to lose weight.
Query: I want a late-night snack.



PDR = 5.88% **RTR = 18%**

Performance: **Rapid Collapse**; fails abruptly as dialogue context grows.